

Townsend 7.1-7.2

Questions for Gallicchio

I didn't quite understand why we have the limitation $\eta \geq 0$, but that seems like a hugely important point - since it tells us all the energy values of SHO, and that there are infinite up and a minimum. The physical importance is huge, but where do we justify it from?

7.1

SHO arise from Taylor expanding functions very close to equilibrium. Derivative is zero, and potential matters relative, so we can ignore offsets - that gives just the

$$\frac{1}{2}kx^2$$

potential term. Everything is a Simple Harmonic Oscillator!

7.2

We introduce something analogous to the raising and lowering operators but with units so that we can combine momentum and position. They are themselves raising and lowering

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p}_x \right)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p}_x \right)$$

Which we can rearrange

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger)$$

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a} - \hat{a}^\dagger)$$

$\hat{N} = \hat{a}^\dagger \hat{a}$ is the "Number Operator".

it has eigenstates $\hat{N}|\eta\rangle = \eta|\eta\rangle$, all of which are integers.
where

$$\hat{a}|\eta\rangle = |\psi\rangle$$

$$\langle\psi|\psi\rangle = \eta\langle\eta|\eta\rangle$$

It is the commutation relations

$$\begin{aligned} [\hat{N}, \hat{a}] &= [\hat{a}^\dagger \hat{a}, \hat{a}] \\ &= [\hat{a}^\dagger, \hat{a}] \hat{a} \\ &= -\hat{a} \end{aligned}$$

and

$$\begin{aligned} [\hat{N}, \hat{a}^\dagger] &= [\hat{a}^\dagger \hat{a}, \hat{a}^\dagger] \\ &= \hat{a}^\dagger [\hat{a}, \hat{a}^\dagger] \\ &= \hat{a}^\dagger \end{aligned}$$

$$\begin{aligned} \hat{N} \hat{a}^\dagger |\eta\rangle &= (\hat{a}^\dagger \hat{N} + \hat{a}^\dagger) |\eta\rangle \\ &= (\hat{a}^\dagger \eta + \hat{a}^\dagger) |\eta\rangle \\ &= (\eta + 1) \hat{a}^\dagger |\eta\rangle \end{aligned}$$

$$\hat{a}^\dagger |\eta\rangle = c_+ |\eta + 1\rangle$$

$$\hat{a} |\eta\rangle = c_- |\eta - 1\rangle$$